Predicting NBA Game Totals Against the Sportsbook Over/Under

UE24CS352A Machine Learning, Mini-Project

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Problem: 'The Bank is Open: AI in Sports Gambling'

1. Problem statement

Before every NBA game, sportsbooks publish an Over/Under line: their estimate of the combined points both teams will score. A bettor wins by correctly guessing whether the real total ends above or below that line. Our task is to predict the total points of each game using only information available before tip-off, to compare our error with the sportsbook's, and to measure how often our predictions pick the correct side of the line. We follow the approach of Bucquet and Sarukkai (Stanford), who used a Random Forest, collaborative filtering, a neural network and an LSTM. We add Ridge regression and longer-history features, and we repeat the test on recent seasons.

2. Dataset

We used the Kaggle 'NBA Betting Data' file, which has 24,440 games from 2007-08 to 2025-26 with final scores, quarter and overtime scores, and the closing Over/Under line (`total`). Its early seasons come from Sportsbook Reviews Online. We kept regular-season games only (22,808). The line is used only to evaluate the models and never as an input feature, as in the paper. Seasons are labelled by the year they end, and we split by season so that no future game is ever used to predict a past one.

Split	Seasons	Games	Used for
First (paper)	2012-13 to 2015-16	4,919	Training
	2016-17	1,230	Validation
	2017-18	1,230	Test
Second (recent)	2020-21 to 2023-24	4,770	Training
	2024-25	1,230	Validation
	2025-26	1,230	Test

The file has no box-score statistics (rebounds, assists, steals, offensive rating) and no travel distance, which the paper used. We replaced them with scoring-based features.

3. Approach

Features. For each game we describe both teams using:

- Their last three games: points scored, points allowed, points in each quarter and overtime points.
- For each of those games, the opponent's season-to-date averages of the same statistics, to reflect how strong the opponent was.
- Each team's average points scored and allowed over its last 10 and last 20 games. This is our addition: the paper's three-game window is very noisy, and a longer history gives a steadier picture of each team.
- Days of rest for each team.

This gives 28 values per game step (both teams, three steps) plus 10 extra values, 94 inputs when flattened. Inputs are standardised using training data only. All models minimise mean squared error (MSE) on the total points.

Models. We compare five models:

- Ridge regression: a linear model with a penalty on large weights (strength 1000).
- Random Forest: 300 decision trees that vote, at least 5 games per leaf.
- Collaborative filtering: the idea behind movie recommendations. We build a home-team by away-team matrix of totals from games in the previous 5 days, fill the gaps with a rank-5 SVD, and read off the entry for the matchup.
- Neural network: fully connected layers of size 500, 100, 20 and 1 with ReLU activations.
- LSTM: one layer with 20 hidden units that reads the three past games in order, with dropout 0.2, followed by a linear layer that also receives the 10 extra features.

4. Implementation

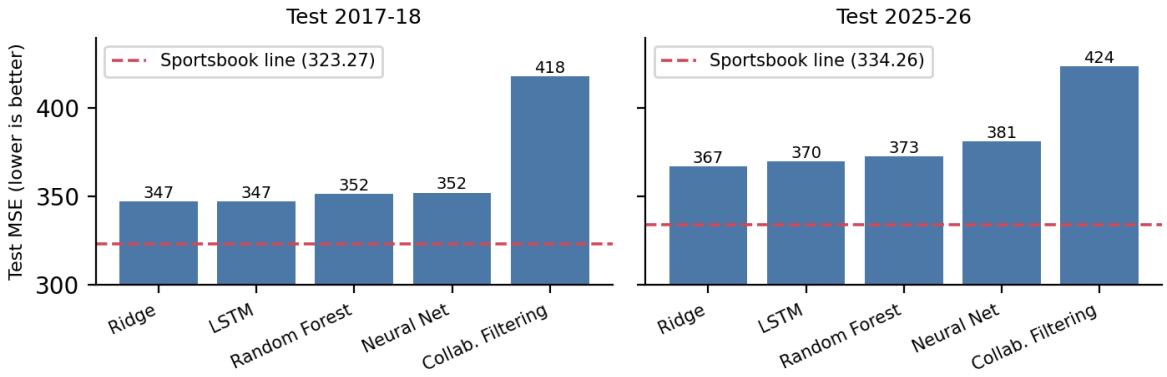
The project is written in Python with NumPy, pandas, scikit-learn and PyTorch. It has four files:

- `prepare_data.py` converts the Kaggle file into a per-team game log and an odds file.
- `features.py` builds the features and the season splits.
- `models.py` holds the five models. The neural network and LSTM use the Adam optimiser with mini-batches of 64, and we keep the epoch with the lowest validation loss.
- `main.py` trains all models and prints the test MSE and the percentage of games where a model picks the correct side of the line (exact ties excluded). The README explains how to set up and run everything.

5. Results

Model	MSE (2017-18)	Beats % (2017-18)	MSE (2025-26)	Beats % (2025-26)
Ridge Regression	347.35	49.2	367.16	49.3
Random Forest	351.73	49.5	372.82	49.2
Collaborative Filtering	418.23	49.6	423.75	50.4
Neural Network	352.30	50.5	381.40	50.2
LSTM	347.36	50.6	370.03	50.9
Sportsbook line	323.27	-	334.26	-

Test season in brackets. 'Beats %' is how often the model picks the correct side (Over or Under) of the sportsbook line.



6. Conclusions

- The sportsbook is the most accurate predictor on both test seasons (MSE 323.3 and 334.3). Our best models, Ridge regression and the LSTM, reach about 347 and 367-370. The differences among our best models are small, so we do not claim one is better.
- Using 10- and 20-game averages cut our error by roughly 10% (the best model with only the last three games reached 385.7), so the paper's short window is too noisy.
- On both splits, all models pick the correct side of the line only 49.2% to 50.9% of the time. On 1,230 games that is within normal chance variation of 50%, and below the 52-53% the paper says is needed to profit after betting fees. We found no reliable betting edge, which agrees with the paper (its sportsbook MSE was 320.7, almost identical to ours).
- Errors are higher on 2025-26 for every model, including the sportsbook, which fits a more variable modern game, but the ranking is unchanged.
- Collaborative filtering was weakest, unlike in the paper, probably because our features are limited and the 5-day window holds few games.

Limitations: no player-level data, injuries or box-score statistics; neural network results can vary slightly across machines. **Future work:** add box-score and player data, and train a classifier that predicts Over/Under directly.